

Part 1:

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following could be valid process states during a process lifetime?

Select one or more:

- ☐ a. jumping
- ☒ b. running ✓
- ☒ c. blocked ✓
- ☐ d. jaberring
- ☒ e. ready ✓
- ☒ f. terminated (like the Linux zombie state) ✓
- ☐ g. flattering

Your answer is correct.

Question **2**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

Which of the following memory allocation strategies suffers by external fragmentation?

Select one or more:

- ☐ a. paging
- ☒ b. segmentation ✓
- ☐ c. swapping
- ☐ d. best fit
- ☒ e. single contiguous area for a process ✓

Your answer is correct.

Question **3**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

Which types of pipes do the Linux and Windows operating systems provide?

Select one or more:

- ☒ a. named ✓
- ☐ b. triangular
- ☒ c. anonymous ✓
- ☐ d. unidirectional
- ☐ e. reader-writer

Your answer is correct.

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

How many threads could run in the same time in the critical region of the function `aeroplane()` in the code below?

```
pthread_mutex_t airport = PTHREAD_MUTEX_INITIALIZER;

void* aeroplane(void* arg)
{
    pthread_mutex_lock(&airport);

    // use a runway to take off or land

    pthread_mutex_unlock(&airport);
}

int main(int argc, char **argv)
{
    pthread_t th[23];

    for (i = 0; i < 23; i++) {
        pthread_create(&th[i], NULL, aeroplane, NULL);
    }

    for (i = 0; i < 23; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Answer: 1

**Question 5**

Partially correct

Mark 0.10 out of 1.00

Flag question

In its role as a *virtual machine provider*, an operating system executes the following actions

Select one or more:

- ☒ a. decides which user process to give a particular processor at one moment ✖
- ☐ b. hides the details and complexity of hardware resources
- ☒ c. shares the system's resources between the competing user programs ✖
- ☒ d. provides a more convenient view of the system, consisting in different abstract concepts and interfaces ✔
- ☐ e. brings the hardware resources in a functional state

Your answer is partially correct.

You have selected too many options.

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

A lock is a mechanisms that allows thread be at one moment in the protected by that lock. It provides the and primitives.

Your answer is correct.

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code of a program creating multiple threads. Identify which threads are CPU-bound and which are I/O bound.

```
#define N 6

void* fc_1(void* arg)
{
    long int steps = (long int) arg;
    int res = 0;

    for (int i = 0; i < 100 * steps; i++) {
        res += pow(res, i) + sqrt(i) * sqrt(i);
    }

    int *val = (int*) malloc(sizeof(int));
    *val = res;
    return val;
}

void* fc_2(void* arg)
{
    long int steps = (long int) arg;
    int res = 0;

    for (int i = 0; i < 1000 * steps; i++) {
        printf("i = %d\n", i);
        res += res + i;
        printf(res = %d\n", res);
    }

    return NULL;
}

int main(void)
{
    pthread th[N];

    for(long int i = 0; i < N; i++) {
        if (i % 2 == 0) {
            pthread_create(&th[i], NULL, fc_2, (void*) i);
        } else {
            pthread_create(&th[i], NULL, fc_1, (void*) i);
        }
    }

    for (long int i = 0; i < N; i++) {
        pthread_join(th[i], NULL);
    }

    return 0;
}
```

th[0] I/O-bound 

```

    *val = res;
    return val;
}

void* fc_2(void* arg)
{
    long int steps = (long int) arg;
    int res = 0;

    for (int i = 0; i < 1000 * steps; i++) {
        printf("i = %d\n", i);
        res += res + i;
        printf(res = %d\n", res);
    }

    return NULL;
}

int main(void)
{
    pthread th[N];

    for(long int i = 0; i < N; i++) {
        if (i % 2 == 0) {
            pthread_create(&th[i], NULL, fc_2, (void*) i);
        } else {
            pthread_create(&th[i], NULL, fc_1, (void*) i);
        }
    }

    for (long int i = 0; i < N; i++) {
        pthread_join(th[i], NULL);
    }

    return 0;
}

```

th[0]	I/O-bound	⬆️	✓
th[2]	I/O-bound	⬆️	✓
th[1]	CPU-bound	⬆️	✓
th[4]	I/O-bound	⬆️	✓
th[5]	CPU-bound	⬆️	✓
th[3]	CPU-bound	⬆️	✓

Your answer is correct.

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

The file system in an OS usually provides a 's contents to user applications as a .

Your answer is correct.

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

Supposing that a computer has 51 (logical) processors, how many threads could be simultaneously in the *running* state?

Answer: 

Question 10

Partially correct

Mark 0.88 out of 1.00

Flag question

Which of the following code sequences from an user-application's could lead to a switch from user to kernel mode?

Select one or more:

☐

```
char *p = NULL;  
*p = 10;
```

☒

```
int fd = creat("file", 0766);
```

☒

```
char buf[10];  
int x = 1;  
n = read(fd, buf, x);
```

☐

```
char buf[10];  
buf[0] = 0;
```

☒

```
write(1, "AAA", 2);
```

☒

```
int pid = fork();
```

☐

```
int x = 1;  
x++;
```

☒

```
printf("Hello world\n");
```

☒

```
char buf[10];  
int x = 10;  
buf[0] = x / 0;
```

☒

```
close(1);
```

☒ `int fd = creat("file", 0766);`



☒ `char buf[10];`
`int x = 1;`
`n = read(fd, buf, x);`



☐ `char buf[10];`
`buf[0] = 0;`

☒ `write(1, "AAA", 2);`



☒ `int pid = fork();`



☐ `int x = 1;`
`x++;`

☒ `printf("Hello world\n");`



☒ `char buf[10];`
`int x = 10;`
`buf[0] = x / 0;`



☒ `close(1);`



Your answer is partially correct.
You have correctly selected 7.

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

Match the number of processes and anonymous pipes created by the shell in order to handle the following command line.

```
ls -l /etc | cat | grep -i "os-exam" | sort
```

no of pipes

3



no of processes

4



Your answer is correct.

Question 12

Partially correct

Mark 0.80 out of 1.00

Flag question

Which memory management strategies use base and limit registers for translating virtual addresses to physical addresses?

Select one or more:

- ☒ a. single contiguous area for a process
- ☐ b. best fit
- ☐ c. swapping
- ☒ d. paging
- ☒ e. segmentation

Your answer is partially correct.

You have selected too many options.

Question 13

Correct

Mark 1.00 out of 1.00

Flag question

Match the following file types to their *access method* the OS usually imposes to user applications over accesses to their contents.

regular files	random access, i.e. access bytes in any order	✓
pipes	sequential method, based on FIFO	✓
directories	sequential method, i.e. traverse the element collection from beginning to end	✓

Your answer is correct.

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

An *operating system* is a special , whose role is to computer's resources and provide the a convenient and safe environment to their in.

Your answer is correct.

Question 15

Incorrect

Mark 0.00 out of 1.00

Flag question

You are given the following code of a program run on an OS using paging for memory management, with pages of 1 KiB (i.e 1024 bytes).

```
int main()
{
    char *p = (char*) malloc(182);
    if (p == NULL)
        exit(1);
    for (int i = 0; i < 182; i++)
        p[i] = i;
    free(p);
}
```

Which is the *amount of memory* (in bytes) that could be lost due to *internal fragmentation*, during the given program runtime?

Answer: 182 ✖

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

An operating system is closely tied to (i.e. dependent by) the hardware it runs on.

Select one:

- ☒ True ✔
- ☐ False

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

How many threads could run in the same time in the critical region of the function `aeroplane()` in the code below?

```
sem_t airport;

void* aeroplane(void* arg)
{
    sem_wait(&airport);

    // use a runway to take off or land

    sem_post(&airport);
}

int main(int argc, char **argv)
{
    pthread_t th[43];

    sem_init(&airport, 0, 8);

    for (i = 0; i < 43; i++) {
        pthread_create(&th[i], NULL, aeroplane, NULL);
    }

    for (i = 0; i < 43; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Answer: 8



Question 18

Correct

Mark 1.00 out of 1.00

Flag question

From the user perspective, a *text file* is a file in which every is a code of a printable and also contains special codes for marking the of a text .

Your answer is correct.

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

A semaphore is a mechanisms that allows threads be at one moment in the protected by that semaphore. It provides the and primitives.

Your answer is correct.

Question 20

Correct

Mark 1.00 out of 1.00

Flag question

Match the following library functions with the system calls that are normally called by those functions.

<code>scanf(...)</code>	read(0, &c, 1);	✓
<code>printf(...)</code>	write(1, buf, strlen(buf));	✓
<code>perror(...)</code>	write(2, buf, strlen(buf));	✓

Your answer is correct.

Question **21**

Correct

Mark 1.00 out of 1.00

🚩 Flag question

Match the following concepts to their correct definition.

thread	a particular execution path in a running program	↕	✓
process	a running a program in a "virtual computer"	↕	✓
directory	a fundamental data organization mechanism	↕	✓
file	the basic storage unit on a file system	↕	✓

Your answer is correct.

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code of a multi-threaded program that will be run on a system with 21 logical processors, an information that could be obtained at runtime by calling the `get_cpus_no()` syscall. Which is the optimum number of threads (i.e. the value of the `NO_OF_THREADS` variable) that should be created by the given program, such that to gain the best performance, i.e. the shortest running time. It is supposed no other program is running in the same time on the given system.

NOTE: your answer must be a number, not a piece of code.

```
unsigned int NO_OF_THREADS;

int *numbers;
size_t n;

void* fc(void* arg)
{
    long int id = (long int) i;

    for (int i = id; i < n; i += NO_OF_THREADS)
        numbers[i] += i * id;

    return NULL;
}

int compute_optimum_no_of_threads()
{
    // TODO: replace with the best value
    return 0;
}

int main(void)
{
    int pthread_t th[1000];

    NO_OF_THREADS = compute_optimum_no_of_threads();

    n = 100000000 * NO_OF_THREADS * sizeof(int);
    numbers = (int*) malloc(n);
    if (numbers == NULL)
        exit(1);

    bzero((char*)numbers, n);

    for(long int i = 0; i < NO_OF_THREADS; i++)
        pthread_create(&th[i], NULL, fc, (void*) i);

    for (long int i = 0; i < NO_OF_THREADS; i++)
        pthread_join(th[i], NULL);

    free(numbers);
}
```

Answer: 21



Question 23

Correct

Mark 1.00 out of 1.00

 Flag question

The two main components of a file in any file systems are:

Select one or more:

- ☐ a. file's parent directory
- ☒ b. the file's data (i.e. contents) ✓
- ☐ c. file's owner
- ☐ d. file's access permissions
- ☒ e. the file's metadata structure ✓

Your answer is correct.

Question 24

Correct

Mark 1.00 out of 1.00

 Flag question

You are given the following code, which creates a file. Supposing the file is created successfully, which of the following actions can be performed successfully on the created file.

```
int fd = create("test", 0651);
```

Select one or more:

- ☒ a. owner can read the file ✓
- ☒ b. owner can write the file ✓
- ☐ c. owner can execute the file
- ☒ d. users in group with owner can read the file ✓
- ☐ e. users in group with owner can write the file
- ☒ f. users in group with owner can execute the file ✓
- ☐ g. other users can read the file
- ☐ h. other users can write the file
- ☒ i. other users can execute the file ✓

Your answer is correct.

Part 2:

Question 1

Correct

Mark 1.00 out of 1.00

🚩 Flag question

How many page table entries are there in a page table of a process on a system using 75 bits for memory addresses and pages of 2^{11} bytes?

NOTE: You must introduce the power of 2 of the computed number, i.e. if it is of the form 2^{10} , your response should be 10.

Answer:

64



Question 2

Correct

Mark 1.00 out of 1.00

Flag question

Indicate for each code sequence given below, supposed to be run by a shell, the handled command line.

1.

```
pid = fork();
if ( pid > 0 ) {
    wait(NULL);
} else {
    fd = open("mouse", O_RDONLY);
    dup2(fd, 0);
    close(fd);
    execlp("cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

2.

```
pid = fork();
if ( pid > 0 ) {
    wait(NULL);
} else {
    fd = open("mouse", O_WRONLY);
    dup2(fd, 1);
    close(fd);
    execlp("cat", "cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

3.

```
pid = fork();
if ( pid > 0 ) {
} else {
    fd = open("mouse", O_WRONLY);
    dup2(fd, 1);
    close(fd);
    execlp("cat", "cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

Your answer is correct.

Question 3

Correct

Mark 1.00 out of 1.00

 Flag question

You are given the following code, which creates a file. Supposing the file is created successfully, which of the following actions can be performed successfully on the created file.

```
umask("0037");  
int fd = create("test", 0651);
```

Select one or more:

- ☒ a. owner can read the file ✓
- ☒ b. owner can write the file ✓
- ☐ c. owner can execute the file
- ☒ d. users in group with owner can read the file ✓
- ☐ e. users in group with owner can write the file
- ☐ f. users in group with owner can execute the file
- ☐ g. other users can read the file
- ☐ h. other users can write the file
- ☐ i. other users can execute the file

Your answer is correct.